Machine Intelligence 1

2.2 Support Vector Machines

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2.2.1 Structural Risk Minimization

A bound on the generalization error

- finite samples: bound on the generalization error (for all $\eta > 0$) (see Chapter 2.1.3)

$$P\left\{ \sup_{\underline{\mathbf{w}} \in \Lambda} \left| R_{[\underline{\mathbf{w}}]} - R^{(p)}_{\mathrm{emp}[\underline{\mathbf{w}}]} \right| > \eta \right\} < \underbrace{4 \exp\left(G^{\Lambda}_{(2p)} - p\left(\eta - \tfrac{1}{p}\right)^2 \right)}_{\stackrel{!}{=}\epsilon}$$

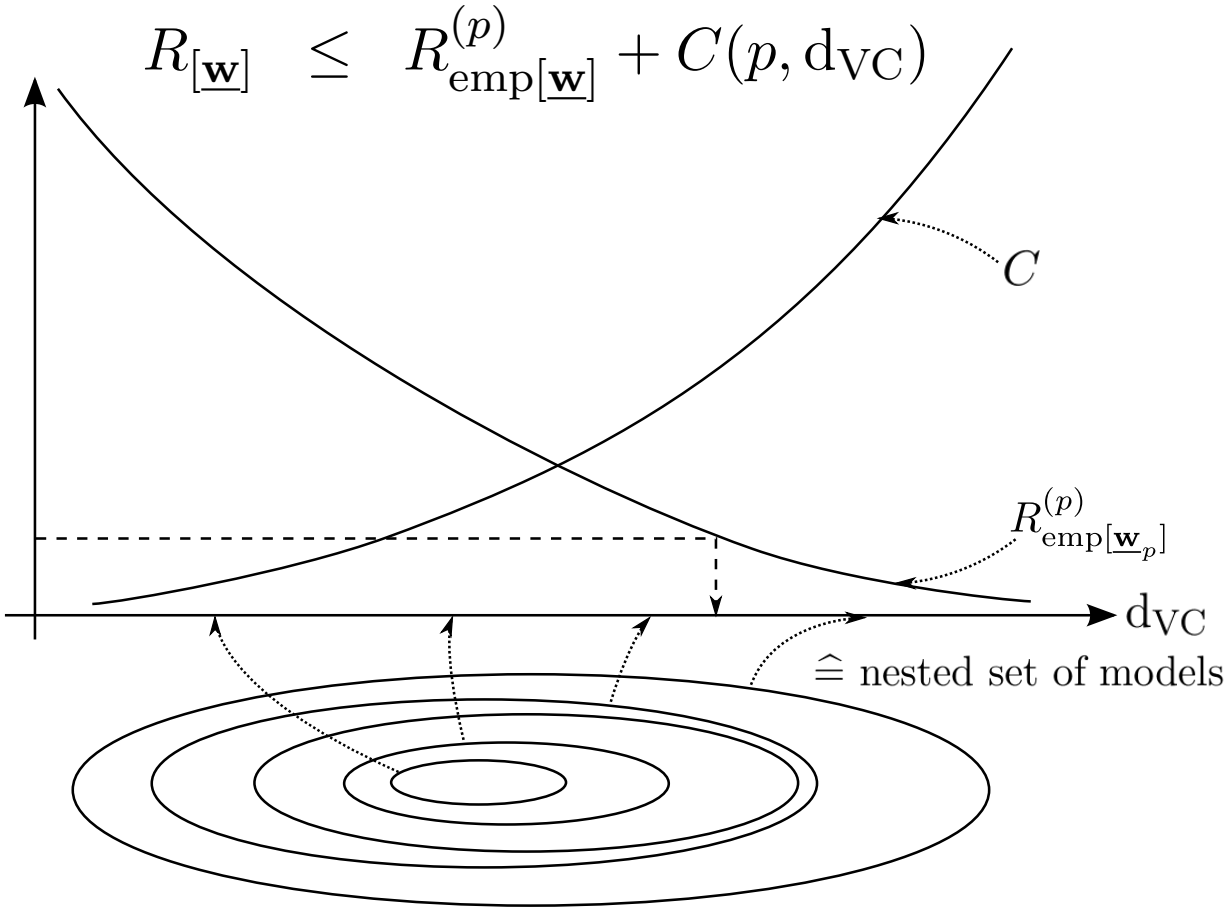
- with probability larger or equal than $1 - \epsilon$ we obtain:

$$R_{[\underline{\mathbf{w}}]} \leq \underbrace{R^{(p)}_{\mathrm{emp}[\underline{\mathbf{w}}]}}_{\text{empirical error}} + \underbrace{\left(\frac{G^{\Lambda}_{(2p)} - \ln \frac{\epsilon}{4}}{p} \right)^{\frac{1}{2}} + \frac{1}{p}}_{\text{complexity term } C}$$

for all models $\underline{\mathbf{w}}$

- If we replace $G^{\lambda}_{(2p)}$ by its upper bound, then, for a given ϵ, the complexity term C depends only on p and d_{VC}.

A bound on the generalization error



underfitting $\leftarrow$ ... appropriate model complexity ... $\rightarrow$ overfitting

Structural Risk Minimization (SRM)

$$R_{[\underline{\mathbf{w}}]} \quad \leq \quad R^{(p)}_{\mathrm{emp}[\underline{\mathbf{w}}]} + C(p, \mathrm{d}_{\mathrm{VC}})$$

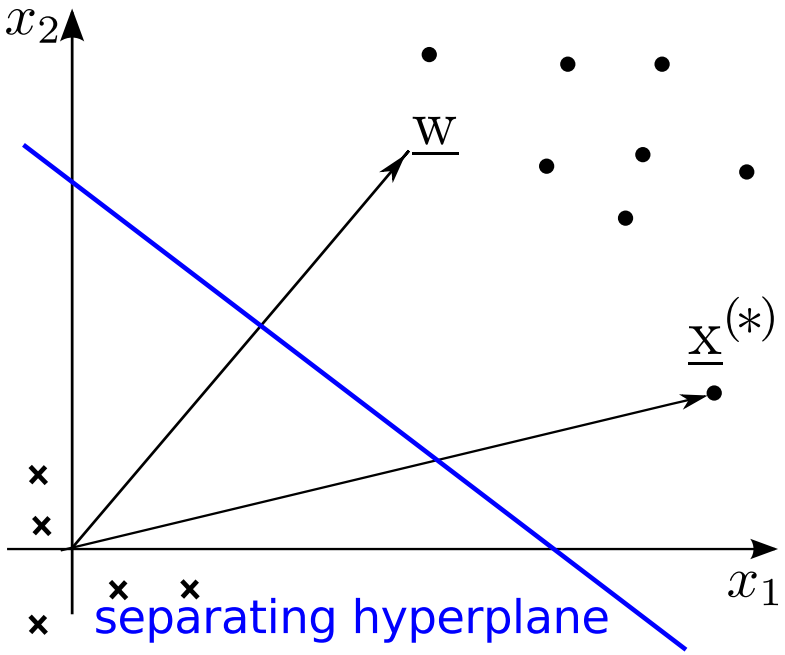
- Minimize complexity $C(p, \mathrm{d}_{\mathrm{VC}})$ of the model class while keeping the empirical error $R^{(p)}_{\mathrm{emp}[\underline{\mathbf{w}}]}$ bounded.
- SRM-learning is consistent *(cf. Vapnik, 1998, chapter 6.3)*.

2.2.2 Perceptrons Revisited

Canonical hyperplanes

- **data representation** binary classification: $\underline{\mathbf{x}} \in \mathbb{R}^N, \quad y_T \in \{-1, +1\}$
- **model class:** connectionist neurons $y = \text{sign}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b\right)$
- parameters of the seperating hyperplane $\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b = 0$ are not unique
  - data dependent normalization

$$\min_{\alpha=1,\ldots,p} \left| \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b \right| \stackrel{!}{=} 1$$



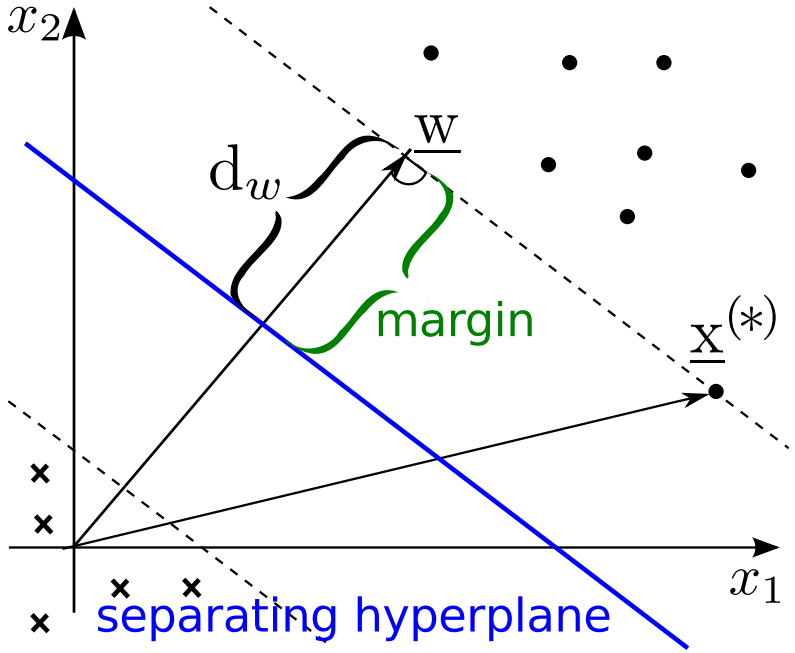
Canonical hyperplanes

- normalized distance to closest point $\underline{\mathbf{x}}^{(*)}$

$$\mathrm{d}_w = \left| \underbrace{\frac{\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(*)}}{\|\underline{\mathbf{w}}\|}}_{\substack{\text{length of the} \\ \text{projection of } \underline{\mathbf{x}} \\ \text{in the direction} \\ \text{of } \underline{\mathbf{w}}}} + \underbrace{\frac{b}{\|\underline{\mathbf{w}}\|}}_{\substack{\text{distance of} \\ \text{the hyperplane} \\ \text{from the origin}}} \right|$$

$$= \frac{1}{\|\underline{\mathbf{w}}\|} \underbrace{\left| \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(*)} + b \right|}_{=1}$$

$$\mathrm{d}_w = \frac{1}{\|\underline{\mathbf{w}}\|}$$

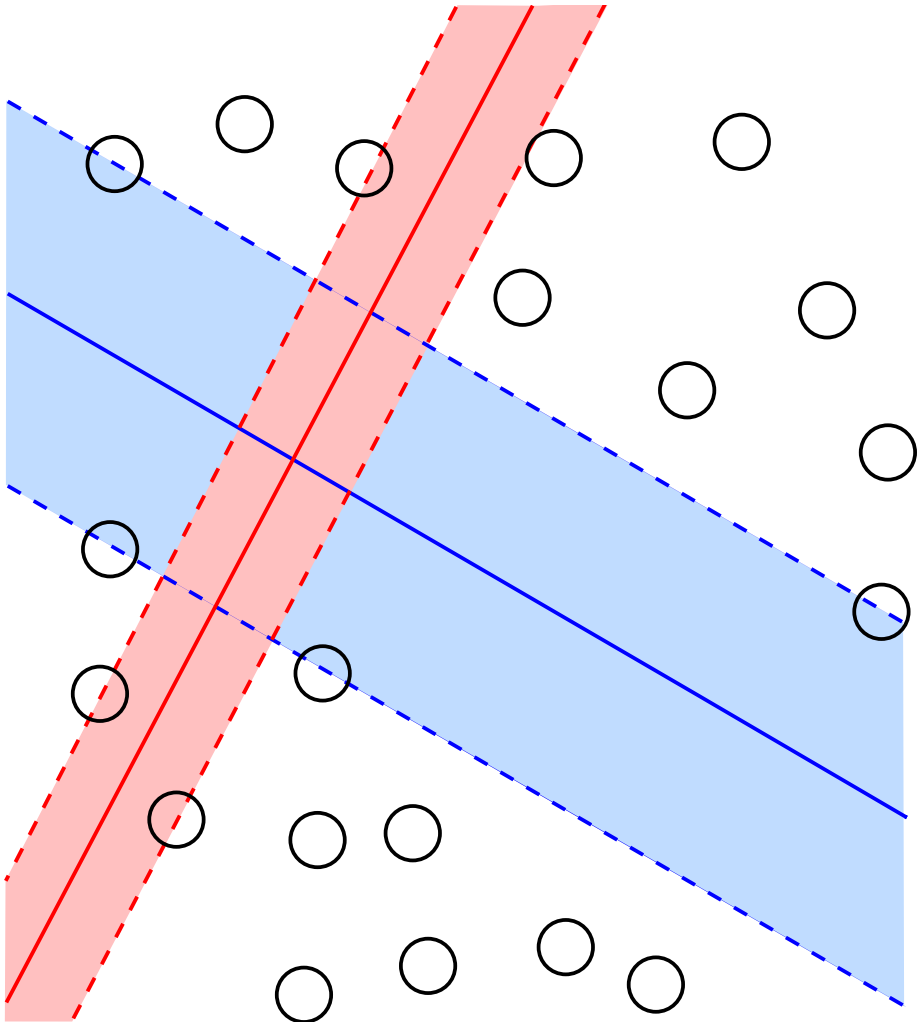


b is negative in the figure as the origin is in the direction of $-\underline{\mathbf{w}}$.

- The minimum normalized distance to the hyperplane is called **margin**.

Margins and the capacity of the model class

Model class: all linear classifiers with a margin larger than a given value d_w



Margins and the VC dimension

Theorem (Vapnik, 1998)

$$\mathrm{d}_{\mathrm{VC}} \quad \leq \quad \min\left(\left\lfloor \frac{\mathrm{d}_R^2}{\Delta_w^2} \right\rfloor, N\right) + 1$$

N : dimension of feature space

Δ_w : lower bound of the margin: $\frac{1}{\|\underline{\mathbf{w}}\|} \geq \Delta_w$

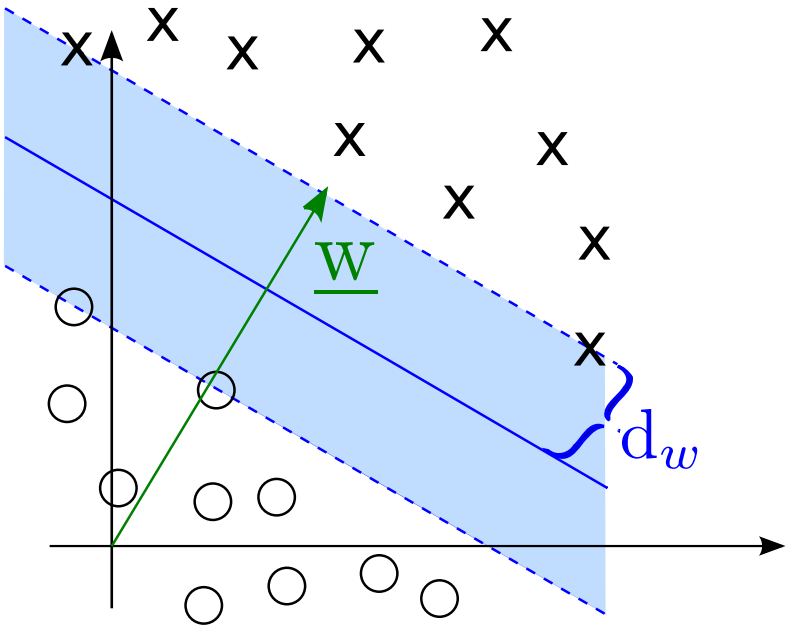
d_R : quantifies the support of $\underline{\mathbf{x}}$: $P(\underline{\mathbf{x}}) \neq 0$ for $\|\underline{\mathbf{x}}\| \leq \mathrm{d}_R$

$\lfloor x \rfloor$: integer part of x

- $\frac{\mathrm{d}_R^2}{\Delta_w^2}$ is independent of the dimension N of feature space.

2.2.3 Learning by Structural Risk Minimization

The primal optimization problem



$$y(\underline{\mathbf{x}}; \underline{\mathbf{w}}) = \mathrm{sign}(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b)$$

$$\mathrm{d}_w = \frac{1}{\|\underline{\mathbf{w}}\|} \stackrel{!}{=} \max$$

$$\tfrac{1}{2}\|\underline{\mathbf{w}}\|^2 \stackrel{!}{=} \min$$

(minimize the capacity...)

s.t. $y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1\,, \quad \forall \alpha\,,$

(... for zero training error and normalized weight vectors)

The method of Lagrange multipliers

$$\underbrace{f_{0(\underline{\mathbf{x}})} \stackrel{!}{=} \min}_{\text{minimization}} \qquad \text{and} \qquad \underbrace{f_{k(\underline{\mathbf{x}})} \leq 0, \quad k = 1, \ldots, m}_{\text{constraints}}$$

$$L_{(\underline{\mathbf{x}},\{\lambda_k\})} \stackrel{!}{=} f_{0(\underline{\mathbf{x}})} + \sum_{k=1}^{m} \lambda_k f_{k(\underline{\mathbf{x}})}\,, \qquad \lambda_k \geq 0\,, \quad \forall k \in \{1, \ldots, m\}$$

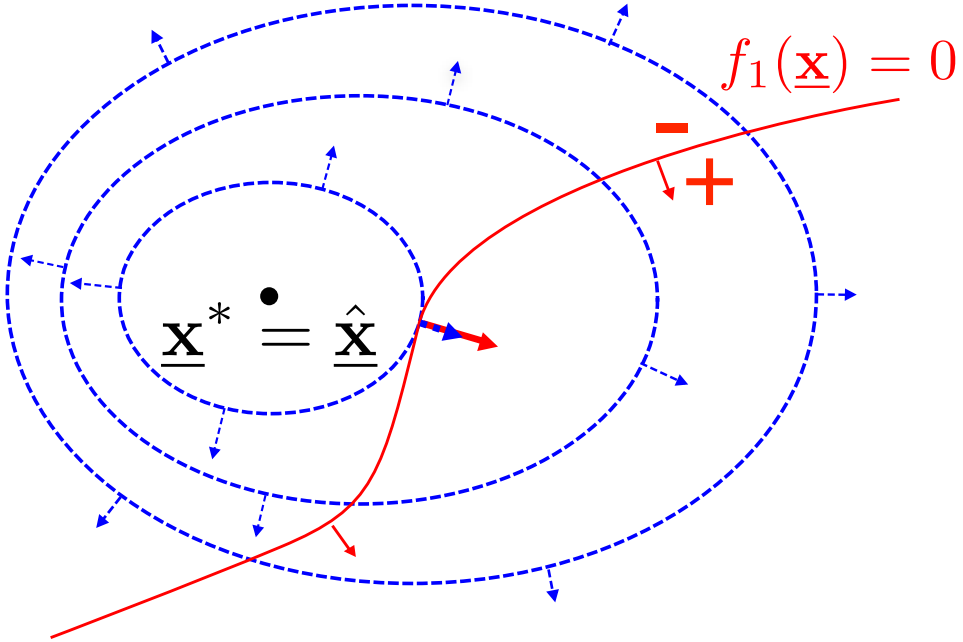
Theorem (Kuhn and Tucker)

Let $A \subset \mathbb{R}^N$ be a convex subset and f_k be convex functions. If there *exists* at least one solution $\underline{\mathbf{x}} \in A$ that satisfies all constrains $f_k(\underline{\mathbf{x}}) \leq 0, \forall k$, then the solution $\underline{\mathbf{x}}^*$ of the constrained optimization problem is given by the saddle point of the Langrangian, i.e.

$$\min_{\underline{\mathbf{x}} \in A} L_{(\underline{\mathbf{x}},\{\lambda_k^*\})} = L_{(\underline{\mathbf{x}}^*,\{\lambda_k^*\})} = \max_{\lambda_k \geq 0} L_{(\underline{\mathbf{x}}^*,\{\lambda_k\})}$$

The method of Lagrange multipliers

$$
\begin{array}{rcll}
f_0(\underline{\mathbf{x}}) & \stackrel{!}{=} & \min & \text{s.t.} \quad f_1(\underline{\mathbf{x}}) \leq 0 \\
L_{(\underline{\mathbf{x}},\lambda_1)} & := & f_0(\underline{\mathbf{x}}) + \lambda_1 f_1(\underline{\mathbf{x}}) & \text{s.t.} \quad \lambda_1 \geq 0
\end{array}
$$



Solution $\underline{\mathbf{x}}^*$ is the unconstrained optimum $\hat{\underline{\mathbf{x}}}$.

- $\frac{\partial L}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} = \frac{\partial f_0}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} + \lambda_1 \frac{\partial f_1}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} = \lambda_1 \frac{\partial f_1}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} = 0 \quad \Rightarrow \quad \lambda_1 = 0$

The method of Lagrange multipliers

$$
\begin{array}{rcll}
f_0(\underline{\mathbf{x}}) & \stackrel{!}{=} & \min & \text{s.t.} \quad f_1(\underline{\mathbf{x}}) \leq 0 \\
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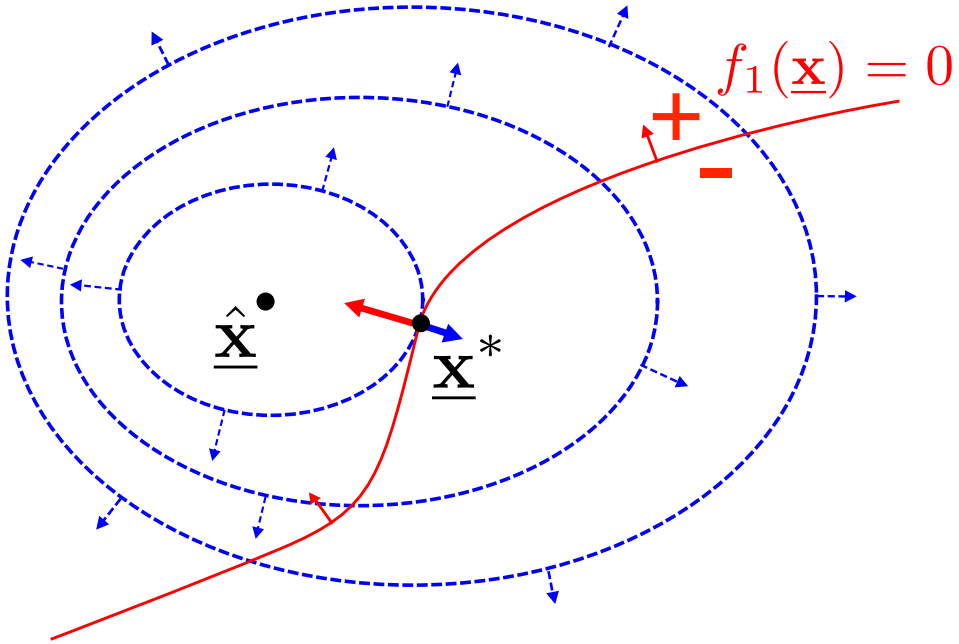
\end{array}
$$



Solution $\underline{\mathbf{x}}^*$ is the constrained optimum (minimum on the constraint surface $f_1(\underline{\mathbf{x}}) = 0$).

- $\frac{\partial L}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} = \frac{\partial f_0}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} + \lambda_1 \frac{\partial f_1}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} = 0 \Rightarrow \frac{\partial f}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} = -\lambda_1 \frac{\partial f_1}{\partial \underline{\mathbf{x}}}\Big|_{\underline{\mathbf{x}}^*} \quad \Rightarrow \quad \lambda_1 \neq 0$

Application to the primal optimization problem

- binary classification with linear connectionist neuron

$$
\begin{aligned}
f_{0(\underline{\mathbf{w}},b)} &= \tfrac{1}{2}\|\underline{\mathbf{w}}\|^2 \\
f_{\alpha(\underline{\mathbf{w}},b)} &= -\Big\{ y_T^{(\alpha)} \Big(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b \Big) - 1 \Big\} \leq 0\,, \quad \forall \alpha \in \{1,\ldots,p\}
\end{aligned}
$$

Lagrangian

$$
L_{(\underline{\mathbf{w}},b,\{\lambda_\alpha\})} = \tfrac{1}{2}\|\underline{\mathbf{w}}\|^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \Big\{ y_T^{(\alpha)} \Big(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b \Big) - 1 \Big\}
$$

$$
\min_{\underline{\mathbf{w}},b} L_{(\underline{\mathbf{w}},b,\{\lambda_\alpha^*\})} = L_{(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha^*\})} = \max_{\lambda_\alpha \geq 0} L_{(\underline{\mathbf{w}}^*,b^*,\{\lambda_\alpha\})}
$$

$\underline{\mathbf{w}}, b$: "primal" variables

λ_α : "dual" variables

(see blackboard)

The dual problem

$$\underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} \lambda_\alpha \, y_T^{(\alpha)} \, \underline{\mathbf{x}}^{(\alpha)}$$

$$L = -\frac{1}{2} \sum_{\alpha,\beta=1}^{p} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top \underline{\mathbf{x}}^{(\beta)}}_{\circledast} + \sum_{\alpha=1}^{p} \lambda_\alpha \stackrel{!}{=} \max_{\{\lambda_\alpha\}}$$

$$\lambda_\alpha \geq 0\,, \quad \forall \alpha \in \{1, \ldots, p\}\,, \qquad \text{and} \qquad \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} = 0 \quad \text{(constraints)}$$

- solved numerically using "sequential minimal optimization" (SMO).

The optimal classifier

- Linear classifier

$$y(\underline{\mathbf{x}}; \underline{\mathbf{w}}) \quad = \quad \text{sign}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b\right)$$

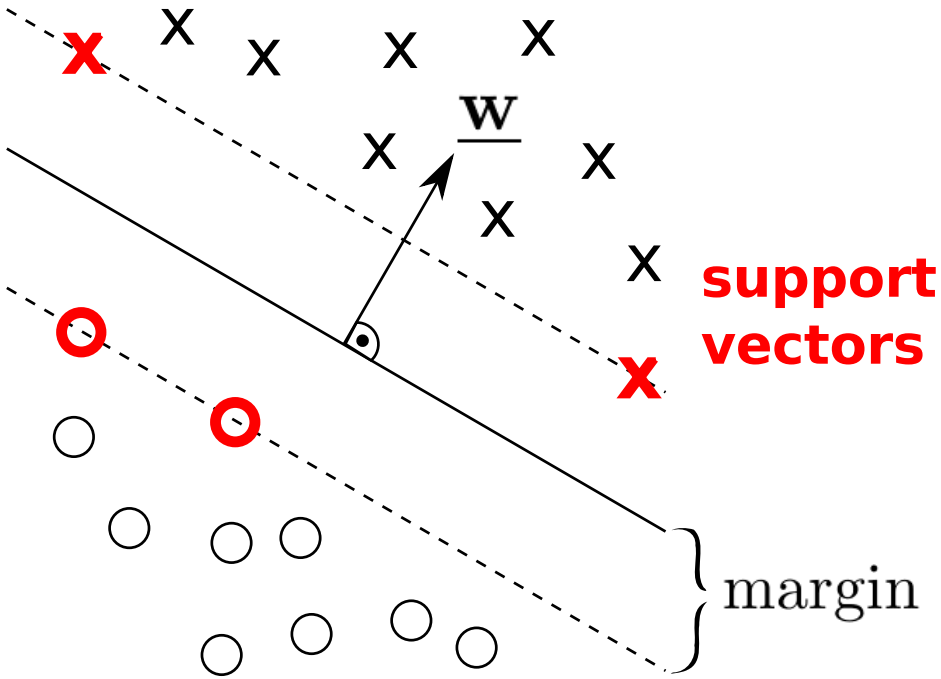
- When $\{\lambda_\alpha^*\}_{\alpha=1}^p$ are known, we can compute

$$\underline{\mathbf{w}}^* \quad = \quad \sum_{\alpha=1}^p \lambda_\alpha^* \, y_T^{(\alpha)} \, \underline{\mathbf{x}}^{(\alpha)} \, ,$$

- and the classifier is

$$y(\underline{\mathbf{x}}) \quad = \quad \text{sign}\left(\sum_{\alpha=1}^p \lambda_\alpha^* \, y_T^{(\alpha)} \underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top \underline{\mathbf{x}}}_{\circledast} + b^*\right) .$$

Support vectors



$\lambda_\alpha \neq 0$ only if:

$$f_\alpha(\underline{\mathbf{w}}^*, b^*) \quad = \quad -\left\{ y_T^{(\alpha)} \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b^* \right) - 1 \right\} \quad \stackrel{!}{=} \quad 0$$

Calculation of the bias

- For all support vectors $\underline{\mathbf{x}}^{(\alpha)}$:

$$\begin{aligned} f_\alpha(\underline{\mathbf{w}}^*, b^*) &= -\left\{ y_T^{(\alpha)} \left(\underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} + b^* \right) - 1 \right\} \quad \stackrel{!}{=} \quad 0 \\ \Rightarrow \quad b^* &= y_T^{(\alpha)} \quad - \quad \underline{\mathbf{w}}^{*\top} \underline{\mathbf{x}}^{(\alpha)} \end{aligned}$$

- The bias b^* is computed by averaging over all support vectors

$$b^* = \frac{1}{\#_{\mathrm{SV}}} \sum_{\alpha \in \mathrm{SV}} \left(y_T^{(\alpha)} - \sum_{\beta \in \mathrm{SV}} \lambda_\beta y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\beta)} \right)^\top \underline{\mathbf{x}}^{(\alpha)}}_{\circledast} \right).$$

Support Vector Machines (SVM)

- **Perceptrons** $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}, b) = \text{sign}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b\right)$ trained by maximizing the margin are called Support Vector Machines (SVMs).

- Computation of Lagrange multipliers:

$$L(\{\lambda_\alpha\}) \;=\; -\frac{1}{2}\sum_{\alpha,\beta=1}^{p} \lambda_\alpha \lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top \underline{\mathbf{x}}^{(\beta)}}_{\circledast} + \sum_{\alpha=1}^{p} \lambda_\alpha \stackrel{!}{=} \max_{\{\lambda_\alpha\}}$$

$$\{\lambda_\alpha \geq 0\}_{\alpha=1}^{p}\,, \quad \textstyle\sum_{\alpha=1}^{p} \lambda_\alpha\, y_T^{(\alpha)} = 0$$

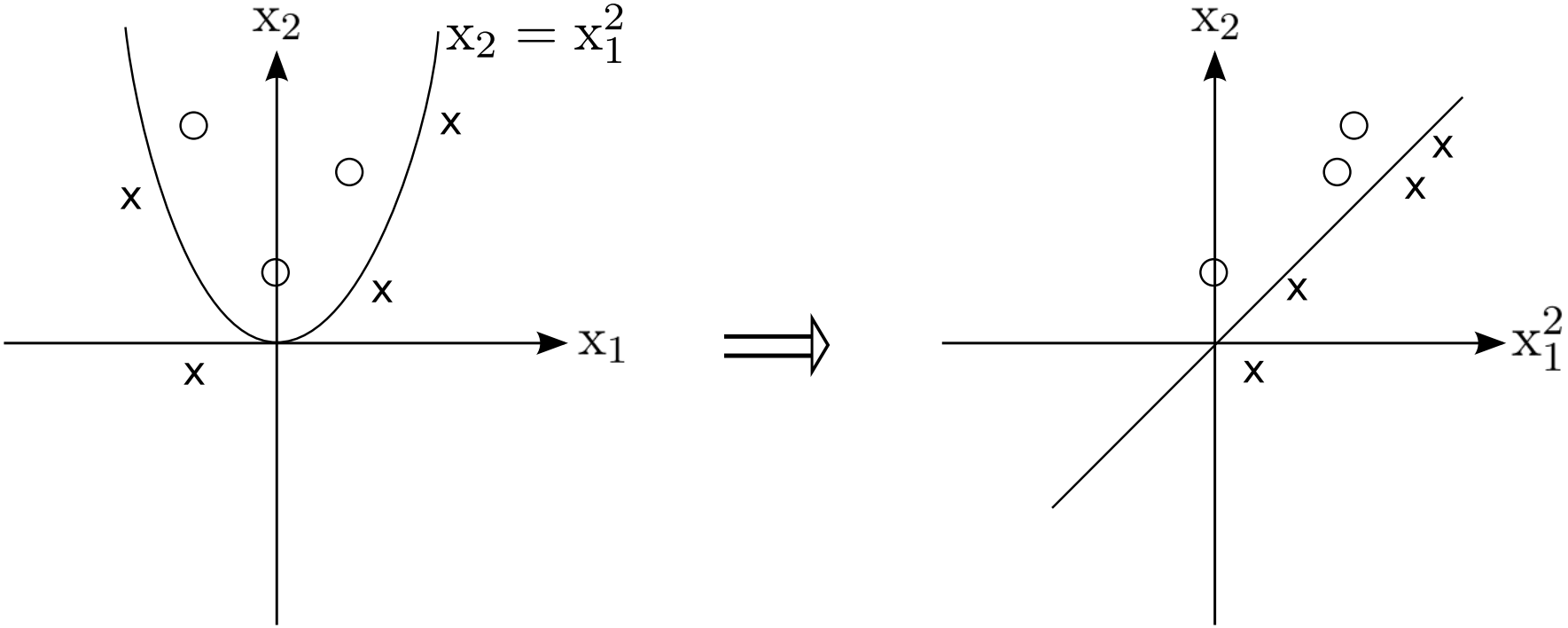
- Construction of the SVM classifier :

$$y(\underline{\mathbf{x}}) \;=\; \text{sign}\left(\sum_{\alpha\in\text{SV}} \lambda_\alpha\, y_T^{(\alpha)} \underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top \underline{\mathbf{x}}}_{\circledast} \;+\; b\right)$$

$$\text{with} \quad b \;=\; \frac{1}{\#_{\text{SV}}} \sum_{\alpha\in\text{SV}} \left(y_T^{(\alpha)} - \sum_{\beta\in\text{SV}} \lambda_\beta y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\beta)}\right)^\top \underline{\mathbf{x}}^{(\alpha)}}_{\circledast}\right)$$

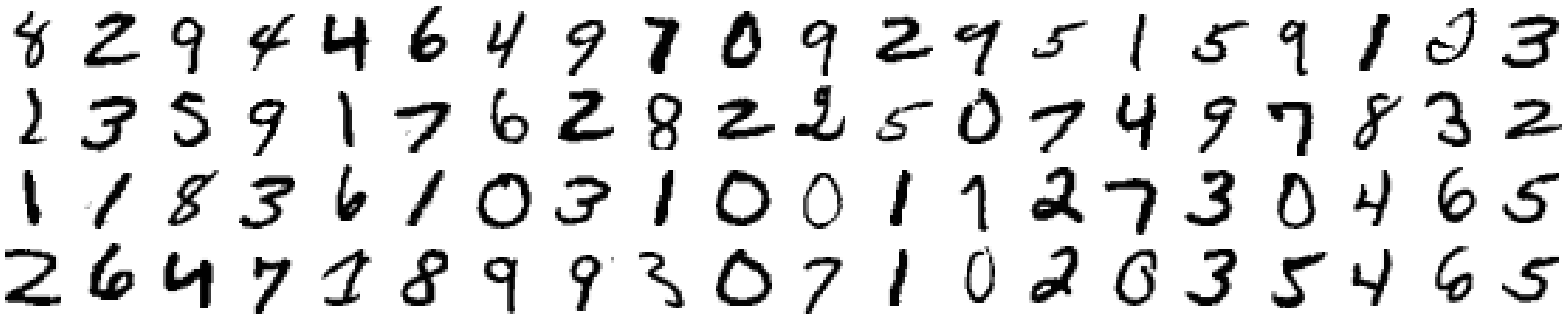
2.2.4 SRM Learning for Non-Linear Classification Boundaries

Transformation of feature space



$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\underline{\mathbf{x}}}_{\text{elementary features}} \longrightarrow \underbrace{\underline{\phi}(\underline{\mathbf{x}})}_{\text{new nonlinear combinations}} = \begin{bmatrix} x_1^2 \\ x_2 \end{bmatrix}$$

Transformation of feature space



- feature space: monomials of degree n

$$\underbrace{\underline{\mathbf{x}}}_{\text{elementary features}} \longrightarrow \underbrace{\underline{\phi}(\underline{\mathbf{x}})}_{\text{monomials of degree } n}$$

- $n = 10$ and N pixel values $x_i \Rightarrow N^{10}$ monomials

The kernel trick

$$\underbrace{\underline{\mathbf{x}}}_{\substack{\text{elementary}\\ \text{features}}} \longrightarrow \underbrace{\underline{\phi}_{(\underline{\mathbf{x}})}}_{\substack{\text{new, nonlinear}\\ \text{features}}}$$

- data enter the SVM formalism only through scalar products $\underline{\phi}^{\top}_{(\underline{\mathbf{x}})}\underline{\phi}_{(\underline{\mathbf{x}})}$ in feature space
- replace scalar products with **kernel function** $\underline{\phi}^{\top}_{(\underline{\mathbf{x}})}\underline{\phi}_{(\underline{\mathbf{x}}')} \rightarrow k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')}$
- e.g. the polynomial kernel $k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = (\underline{\mathbf{x}}^{\top}\underline{\mathbf{x}}' + 1)^n$ is a scalar product in the space of all monomials of degree n

Mercer's theorem

- let $\mathcal{X}$ be a *compact* subset of $\mathbb{R}^N$
- let $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a continuous symmetric function ("kernel")
- let $T_k : L_{2(\mathcal{X})} \to L_{2(\mathcal{X})}$ be the linear convolution operator

$$T_k[f]_{(\underline{\mathbf{x}})} \quad := \quad \int_{\mathcal{X}} k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} f_{(\underline{\mathbf{x}}')} \, d\underline{\mathbf{x}}'$$

- let $\lambda_i \in \mathbb{R}$ be the *eigenvalues* and let $\psi_{i(\underline{\mathbf{x}})} \in L_{2(\mathcal{X})}$ be the *eigenfunctions* of T_k

Mercer's theorem

If T_k is **positive semi-definite**, then the kernel k can be written as a sum

$$k(\underline{\mathbf{x}}, \underline{\mathbf{x}}') \quad = \quad \sum_{i=1}^{\infty} \lambda_i \, \psi_{i(\underline{\mathbf{x}})} \, \psi_{i(\underline{\mathbf{x}}')}$$

of at most countably many terms, where the λ_i are non-negative, and the convergence is absolute and uniform.

Kernel properties

symmetric kernels k

orthonormal eigenfunctions:

$$\int_{\mathcal{X}} \psi_{i(\underline{\mathbf{x}})}\, \psi_{j(\underline{\mathbf{x}})}\, d\underline{\mathbf{x}} \quad = \quad \delta_{ij}, \qquad \forall i, j \in \mathbb{N}$$

positive semi-definite operators T_k

$$\iint\limits_{\mathcal{X}\times\mathcal{X}} k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} f_{(\underline{\mathbf{x}})} f_{(\underline{\mathbf{x}}')}\, d\underline{\mathbf{x}}\, d\underline{\mathbf{x}}' \geq 0\,, \quad \forall f \in L_{2(\mathcal{X})}$$

all eigenvalues λ_i are **non-negative**

Induced feature space

$$k(\underline{\mathbf{x}}, \underline{\mathbf{x}}') \quad = \quad \sum_{i=1}^{\infty} \underbrace{\psi_{i(\underline{\mathbf{x}})} \sqrt{\lambda_i}}_{\phi_{i(\underline{\mathbf{x}})}} \underbrace{\sqrt{\lambda_i}\, \psi_{i(\underline{\mathbf{x}}')}}_{\phi_{i(\underline{\mathbf{x}}')}} \quad = \quad \underline{\phi}^{\top}_{(\underline{\mathbf{x}})} \underline{\phi}_{(\underline{\mathbf{x}}')}$$

- If T_k is positive semi-definite, the kernel k corresponds to an **inner product** in the induced space Φ which is spanned by the features $\phi_{i(\underline{\mathbf{x}})} = \sqrt{\lambda_i}\, \psi_{i(\underline{\mathbf{x}})}$
- Φ is often high dimensional or even of infinite dimension
- a linear classifier in Φ can solve problems which are not linearly separable in the original feature space $\mathcal{X}$

Typical kernel functions

$$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \left(\underline{\mathbf{x}}^\top \underline{\mathbf{x}}' + 1\right)^d$$

polynomial kernel of degree d

$\rightarrow$ feature space Φ spanned by all monomials up to degree d

$\rightarrow$ image processing: pixel correlations

Typical kernel functions

$$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \left(\underline{\mathbf{x}}^\top \underline{\mathbf{x}}' + 1\right)^d$$

polynomial kernel of degree d
$\rightarrow$ feature space Φ spanned by all monomials up to degree d
$\rightarrow$ image processing: pixel correlations

$$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \exp\left\{ - \frac{(\underline{\mathbf{x}}-\underline{\mathbf{x}}')^2}{2\sigma^2} \right\}$$

RBF-kernel with range σ
$\rightarrow$ infinite dimensional feature space

Typical kernel functions

$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \left(\underline{\mathbf{x}}^\top \underline{\mathbf{x}}' + 1\right)^d$

polynomial kernel of degree d
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$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \exp\left\{ - \frac{(\underline{\mathbf{x}}-\underline{\mathbf{x}}')^2}{2\sigma^2} \right\}$

RBF-kernel with range σ
$\rightarrow$ infinite dimensional feature space

$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \tanh\left\{\kappa \underline{\mathbf{x}}^\top \underline{\mathbf{x}}' + \theta\right\}$

neural network kernel with parameters κ and θ $\rightarrow$ not positive definite

Typical kernel functions

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polynomial kernel of degree d
$\rightarrow$ feature space Φ spanned by all monomials up to degree d
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RBF-kernel with range σ
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neural network kernel with parameters κ and θ $\rightarrow$ not positive definite

$k_{(\underline{\mathbf{x}},\underline{\mathbf{x}}')} = \frac{1}{(\|\underline{\mathbf{x}}-\underline{\mathbf{x}}'\|^2+\epsilon^2)^{N/2}}$

Plummer kernel with parameter ϵ
$\rightarrow$ scale invariant kernel

Comments: applicability of the "kernel trick"

- Mercer's theorem can be used to "kernelize" many different linear methods, both supervised or unsupervised.
  - Fisher discriminant analysis
  - principal component analysis *(see MI 2)*
  - k-means clustering & self-organizing maps
  - canonical correlation analysis

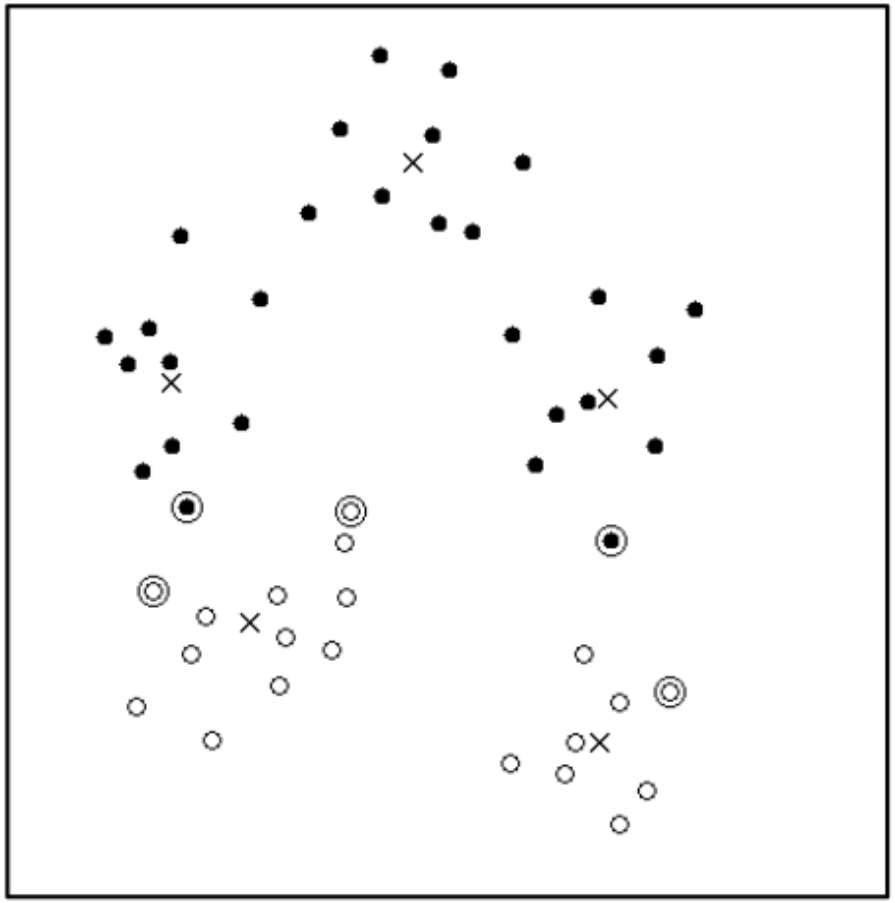
Comments: SVM vs. RBF networks

- RBF network for classification*
  - 5 Gaussian basis functions (centroids $\times$):

$$y(\underline{\mathbf{x}}) = \text{sign}\left(\sum_{i=1}^{5} w_i \exp\left(-\frac{1}{2\sigma_i^2}\|\underline{\mathbf{x}} - \underline{\mathbf{t}}_i\|^2\right)\right)$$

- SVM with Gaussian kernel
  - 5 support vectors $\underline{\mathbf{x}}_i$ ($\circ$)
  - $a_i = \lambda_\alpha y_T^{(\alpha)}$, for $\underline{\mathbf{x}}_i = \underline{\mathbf{x}}^{(\alpha)}$

$$y(\underline{\mathbf{x}}) = \text{sign}\left(\sum_{i=1}^{5} a_i \exp\left(-\frac{1}{2\sigma^2}\|\underline{\mathbf{x}} - \underline{\mathbf{x}}_i\|^2\right) + b\right)$$



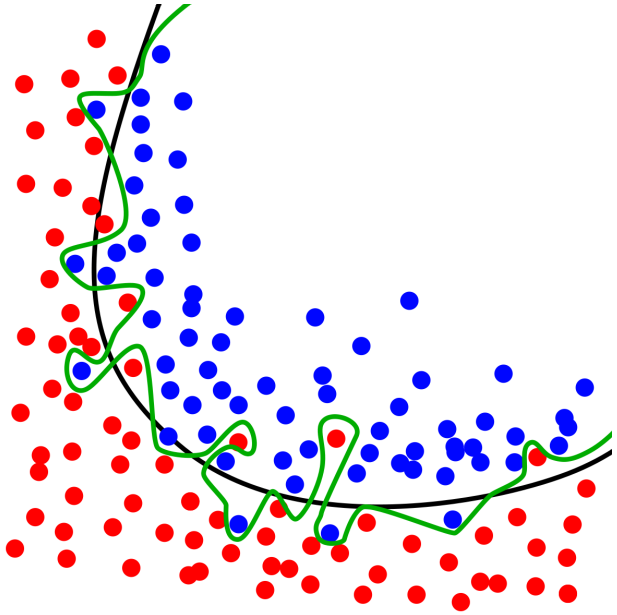
see Schölkopf et al. (1997), Schölkopf and Smola, 2001, p. 204

*Trained by the two-step procedure. Weights were determined by minimizing the squared error for a linear transfer function.

2.2.5 The C-Support Vector Machine

Classification of non-separable problems

- real-world problems are typically non-separable
- incomplete feature sets & noise
- perfect separation of the training set $\rightsquigarrow$ overfitting



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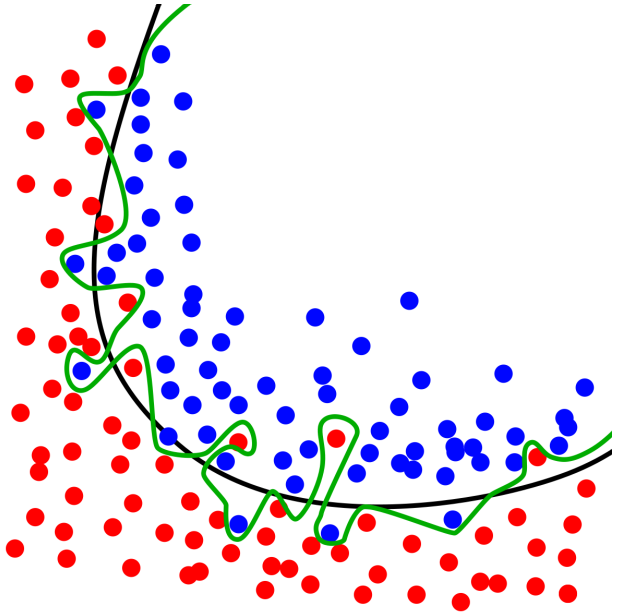
- incomplete feature sets & noise
- perfect separation of the training set $\rightsquigarrow$ overfitting



consequences

$$R_{[\underline{\mathbf{w}}]} \quad \leq \quad R^{(p)}_{\mathrm{emp}[\underline{\mathbf{w}}]} \quad + \quad C(p, \mathrm{d_{VC}})$$

- finite training error $R^{(p)}_{\mathrm{emp}} \neq 0$
- trade-off between the minimization of the training error and the capacity of the model class

The primal problem

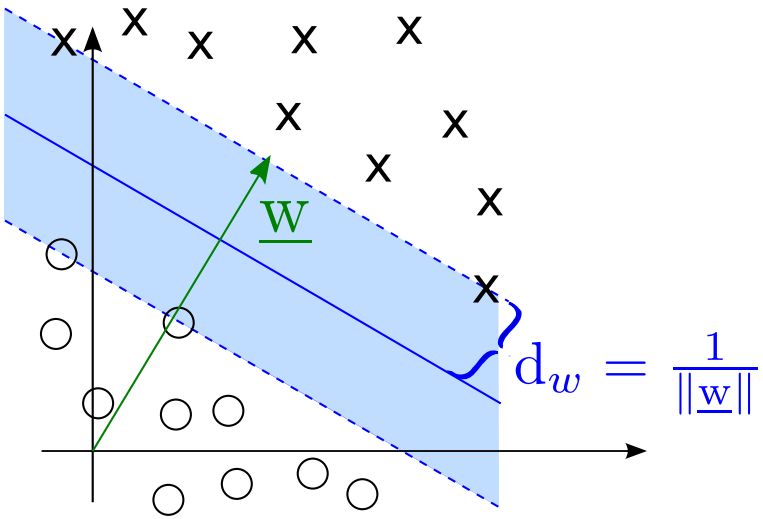
$$\frac{1}{2}\|\underline{\mathbf{w}}\|^2 \quad \stackrel{!}{=} \min \quad \Bigg\{$$

minimize upper bound on VC dimension

constraints ($\forall\alpha$):

$$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\right) \quad \geq 1$$

normalization & correct classification of all data points



The primal problem

$$\frac{1}{2}\|\underline{\mathbf{w}}\|^2 + \frac{C}{p}\sum_{\alpha=1}^{p}\varphi_\alpha \stackrel{!}{=} \min$$

minimize upper bound on VC dimension

+ minimize (approx.) margin error

constraints ($\forall\alpha$):

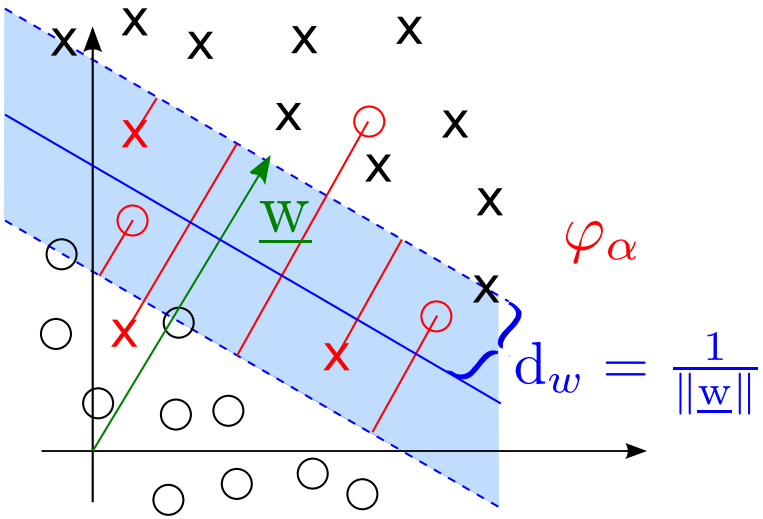
(C: regularization parameter)

$$y_T^{(\alpha)}\left(\underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)} + b\right) \geq 1 - \varphi_\alpha$$

normalization & correct classification of all data points for $\varphi_\alpha = 0$

$$\varphi_\alpha \geq 0$$

"margin errors" for $\varphi_\alpha \neq 0$



Dual problem of the C-SVM

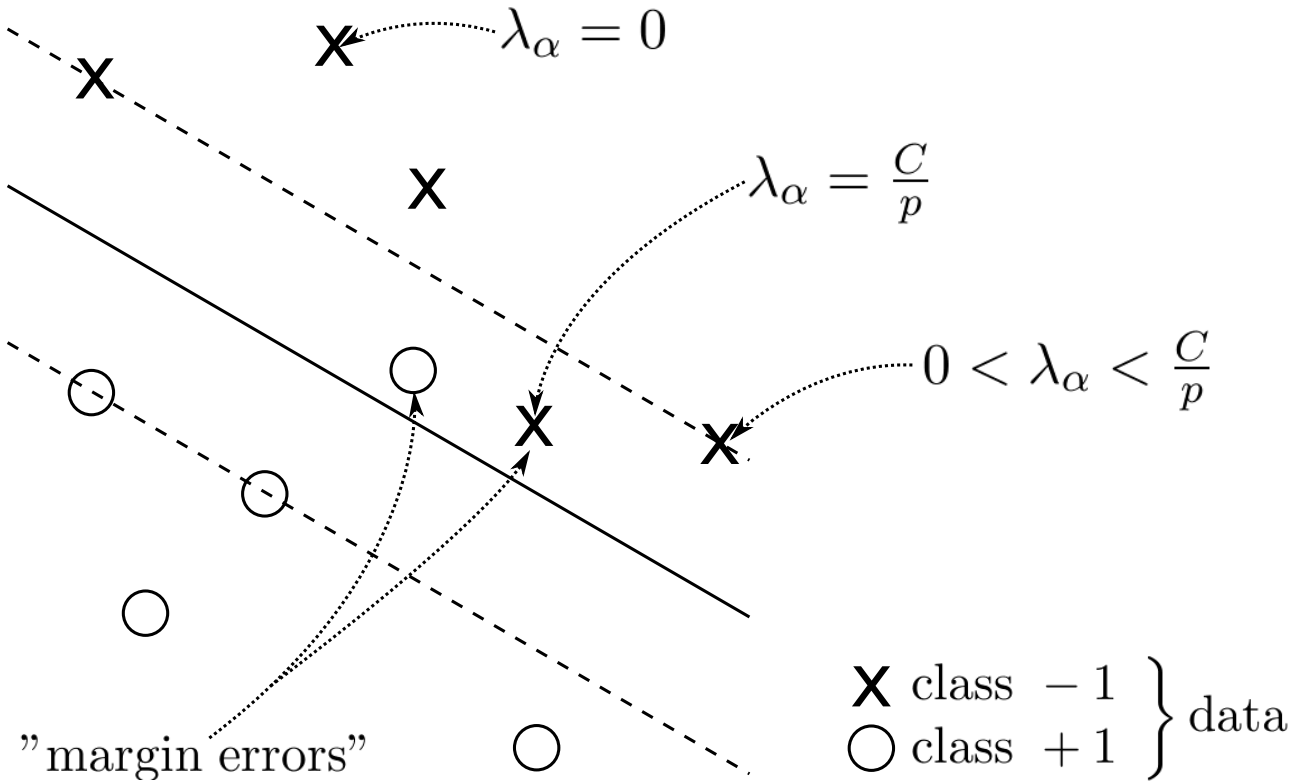
- Objective

$$-\frac{1}{2}\sum_{\alpha,\beta=1}^{p}\lambda_\alpha\lambda_\beta y_T^{(\alpha)}y_T^{(\beta)}\underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top\underline{\mathbf{x}}^{(\beta)}}_{\text{kernel function}}+\sum_{\alpha=1}^{p}\lambda_\alpha \quad \stackrel{!}{=} \quad \max_{\{\lambda_\alpha\}_{\alpha=1}^{p}}$$

- Constraints:

$$\sum_{\alpha=1}^{p}\lambda_\alpha y_T^{(\alpha)}=0 \qquad\qquad 0\leq \underbrace{\lambda_\alpha \leq \frac{C}{p}}_{\text{difference to separable case}}$$

Margin and support vectors



The C-SVM classifier

$$\underline{\mathbf{w}} = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \rightsquigarrow \lambda_\alpha \neq 0 \text{ only for support vectors } SV$$

The C-SVM classifier

$$\underline{\mathbf{w}} = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \rightsquigarrow \lambda_\alpha \neq 0 \text{ only for support vectors } SV$$

$$b = \frac{1}{\#SV_<} \sum_{\alpha \in SV_<} \left(y_T^{(\alpha)} - \sum_{\beta \in SV} \lambda_\beta y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\beta)}\right)^\top \underline{\mathbf{x}}^{(\alpha)}}_{\text{kernel function}} \right)$$

$SV_<$: SVs with $0 < \lambda_\alpha < \frac{C}{p}$ (SVs on the margin)

The C-SVM classifier

$$\underline{\mathbf{w}} = \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)} \qquad \rightsquigarrow \lambda_\alpha \neq 0 \text{ only for support vectors } SV$$

$$b = \frac{1}{\#SV_<} \sum_{\alpha \in SV_<} \left(y_T^{(\alpha)} - \sum_{\beta \in SV} \lambda_\beta y_T^{(\beta)} \underbrace{\left(\underline{\mathbf{x}}^{(\beta)}\right)^\top \underline{\mathbf{x}}^{(\alpha)}}_{\text{kernel function}} \right)$$

$SV_<$: SVs with $0 < \lambda_\alpha < \frac{C}{p}$ (SVs on the margin)

Classifier

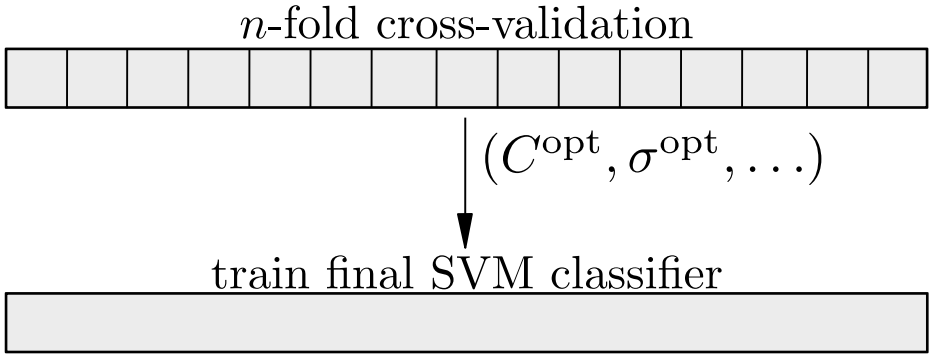
$$y(\underline{\mathbf{x}}) = \text{sign}\left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b\right) = \text{sign}\left(\sum_{\alpha \in SV} \lambda_\alpha y_T^{(\alpha)} \underbrace{\left(\underline{\mathbf{x}}^{(\alpha)}\right)^\top \underline{\mathbf{x}}}_{\text{kernel function}} + b \right)$$

Validation & selection of hyperparameters

- validation and model selection w.r.t. **0-1 loss**

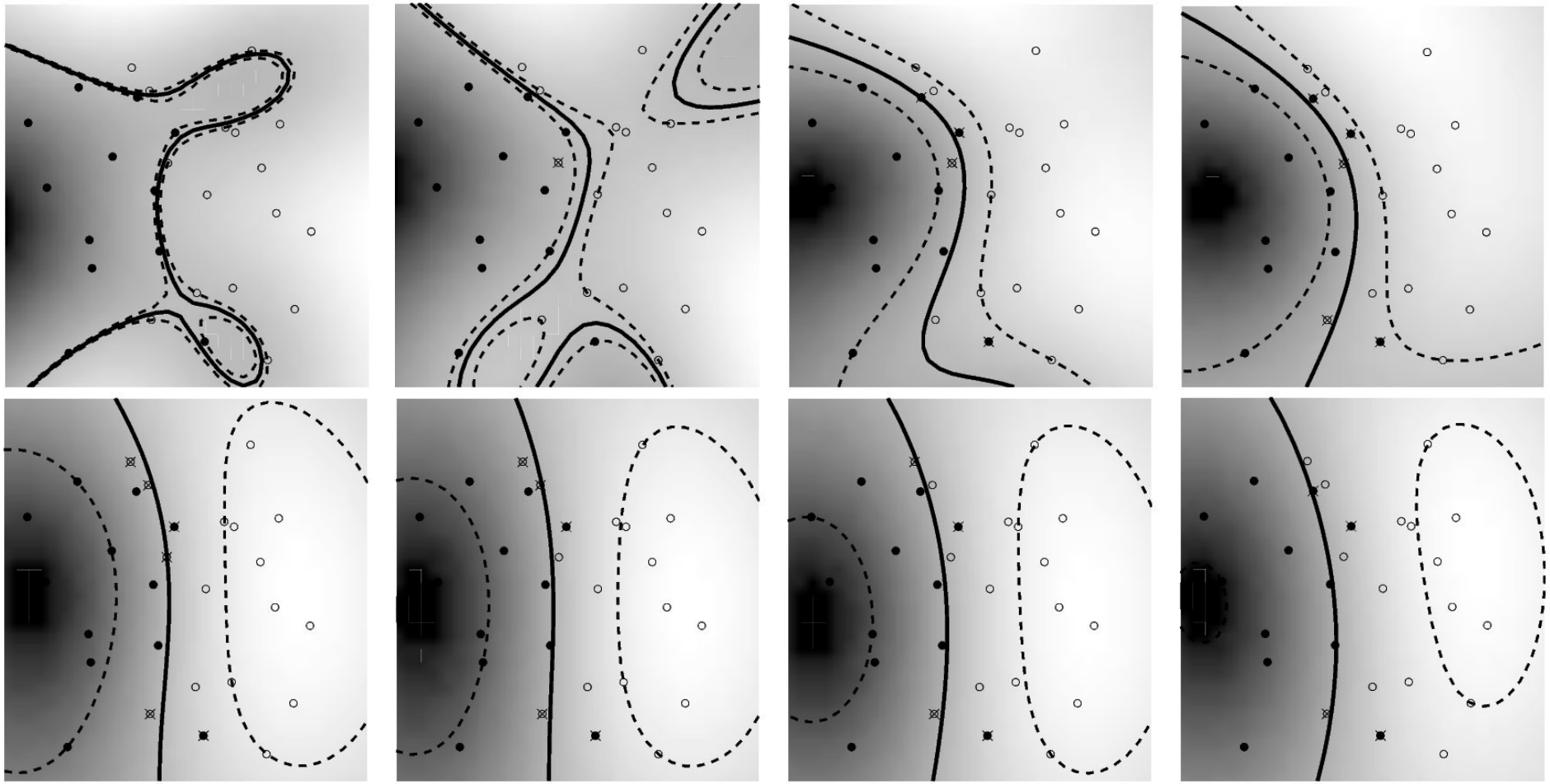
$$e(\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)}) \quad = \quad \begin{cases} 0 & \text{, if } y(\underline{\mathbf{x}}^{(\alpha)}) = y_T^{(\alpha)} \\ 1 & \text{, otherwise} \end{cases}$$

- hyper-parameter selection $(C, \sigma, \ldots)$ by n-fold **cross-validation**



- validation on hold-out **validation set**

SVM and overfitting



(related ν-SVM with $\nu \in \{0.1, 0.2, \ldots, 0.8\}$ and RBF kernel)

see Schölkopf and Smola, 2001, p. 207

2.2.6 Sequential Minimal Optimization

The dual problem

$$-\frac{1}{2}\sum_{\alpha,\beta=1}^{p}\lambda_\alpha\lambda_\beta y_T^{(\alpha)}y_T^{(\beta)}k_{(\underline{\mathbf{x}}^{(\alpha)},\underline{\mathbf{x}}^{(\beta)})}+\sum_{\alpha=1}^{p}\lambda_\alpha \quad \stackrel{!}{=} \quad \max_{\{\lambda_\alpha\}_{\alpha=1}^{p}}$$

$$\text{s.t.} \qquad \sum_{\alpha=1}^{p}\lambda_\alpha y_T^{(\alpha)}=0\,, \qquad\qquad 0\leq \lambda_\alpha \leq \frac{C}{p}\,.$$

The dual problem

$$-\frac{1}{2}\sum_{\alpha,\beta=1}^{p}\lambda_\alpha\lambda_\beta y_T^{(\alpha)}y_T^{(\beta)}K_{\alpha\beta}+\sum_{\alpha=1}^{p}\lambda_\alpha \quad \stackrel{!}{=} \quad \max_{\{\lambda_\alpha\}_{\alpha=1}^{p}}$$

$$\text{s.t.} \qquad \sum_{\alpha=1}^{p}\lambda_\alpha y_T^{(\alpha)}=0\,, \qquad\qquad 0\leq \lambda_\alpha \leq \frac{C}{p}\,.$$

The Gram matrix $\underline{\mathbf{K}}$

$$K_{\alpha\beta}=k_{(\underline{\mathbf{x}}^{(\alpha)},\underline{\mathbf{x}}^{(\beta)})}$$

	1	2	3	...	j
1	K_{11}	K_{12}	...	...	K_{1j}
2	$\vdots$	$\vdots$	K_{23}	...	K_{2j}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i	K_{i1}	K_{i2}	...	...	K_{ij}

The dual problem

$$-\frac{1}{2}\sum_{\alpha,\beta=1}^{p}\lambda_\alpha\lambda_\beta y_T^{(\alpha)}y_T^{(\beta)}K_{\alpha\beta}+\sum_{\alpha=1}^{p}\lambda_\alpha \quad \stackrel{!}{=} \quad \max_{\{\lambda_\alpha\}_{\alpha=1}^{p}}$$

$$\text{s.t.} \qquad \sum_{\alpha=1}^{p}\lambda_\alpha y_T^{(\alpha)}=0\,, \qquad\qquad 0\leq \lambda_\alpha \leq \frac{C}{p}\,.$$

- SVMs operate on pairwise (similarity) data!
- Positive definite kernel $\rightarrow$ positive definite Gram matrix $\underline{\mathbf{K}}$
  $\Rightarrow$ well defined optimization problem.
- $\underline{\mathbf{K}}$ should be pre-computed to speed up subsequent computations.

The SMO procedure

$$-\frac{1}{2}\sum_{\alpha,\beta=1}^{p}\lambda_\alpha\lambda_\beta y_T^{(\alpha)}y_T^{(\beta)}K_{\alpha\beta}+\sum_{\alpha=1}^{p}\lambda_\alpha \quad \stackrel{!}{=} \quad \max_{\{\lambda_\alpha\}_{\alpha=1}^{p}}$$

$$\text{s.t.} \qquad \sum_{\alpha=1}^{p}\lambda_\alpha y_T^{(\alpha)}=0\,, \qquad\qquad 0\leq \lambda_\alpha \leq \frac{C}{p}\,.$$

while *not converged* **do**

Choose two Lagrange multipliers $\lambda_\gamma, \lambda_\delta$.

Optimize the constrained Lagrangian while changing only λ_γ and λ_δ.

end

Choice of λ_γ and λ_δ

Karush-Kuhn-Tucker conditions (KKT conditions)

$$\underbrace{\left[y_T^{(\alpha)} \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b \right) - 1 + \varphi_\alpha \right]}_{\substack{\text{constraint of the primal problem:} \\ = 0 \text{ for all data points on and} \\ \text{within the margin}}} \cdot \underbrace{\lambda_\alpha}_{\substack{= 0 \text{ for all data} \\ \text{points outside} \\ \text{the margin}}} = 0 \qquad \text{(KKT)}$$

1. loop over all λ_γ violating KKT-conditions
   (and additional "threshold"-conditions due to errors in b)
   pick λ_γ for which KKT $\neq 0$
2. for this λ_γ: select a λ_δ which also violates the KKT-conditions and which ensures a "large step" towards the optimum,
   common heuristics: difference in relative errors
   $\left(\sum_{\alpha \in SV} \lambda_\alpha y_T^{(\alpha)} (\underline{\mathbf{x}}^{(\alpha)})^T \underline{\mathbf{x}}^{(\delta)} + b \right) - y_T^{(\delta)}$ vs. $\left(\sum_{\alpha \in SV} \lambda_\alpha y_T^{(\alpha)} (\underline{\mathbf{x}}^{(\alpha)})^T \underline{\mathbf{x}}^{(\gamma)} + b \right) - y_T^{(\gamma)}$
   If r.h.s. is negative, choose data point $\underline{\mathbf{x}}^{(\delta)}$ with positive l.h.s., and vice versa

Reduced optimization problem

$$\min_{\{\lambda_\alpha\}_{\alpha=1}^p} \quad \stackrel{!}{=} \quad \frac{1}{2}\sum_{\alpha\beta}\lambda_\alpha\lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} K_{\alpha\beta} - \sum_\alpha \lambda_\alpha$$

Reduced optimization problem

$$\min_{\{\lambda_\alpha\}_{\alpha=1}^p} \stackrel{!}{=} \frac{1}{2}\sum_{\alpha\beta}\lambda_\alpha\lambda_\beta y_T^{(\alpha)} y_T^{(\beta)} K_{\alpha\beta} - \sum_\alpha \lambda_\alpha$$

$$\min_{(\lambda_\delta,\lambda_\gamma)} \stackrel{!}{=} \frac{1}{2}\Bigg[\lambda_\gamma^2 \overbrace{\underbrace{\left(y_T^{(\gamma)}\right)^2}_{=1} K_{\gamma\gamma}}^{Q_{\gamma\gamma}} + \lambda_\delta^2 \overbrace{\underbrace{\left(y_T^{(\delta)}\right)^2}_{=1} K_{\delta\delta}}^{Q_{\delta\delta}} + 2\lambda_\gamma\lambda_\delta \overbrace{y_T^{(\gamma)} y_T^{(\delta)} K_{\gamma\delta}}^{Q_{\gamma\delta}}\Bigg]$$

$$+ \lambda_\gamma \underbrace{\left[\sum_{\beta\neq(\delta,\gamma)} \lambda_\beta y_T^{(\gamma)} y_T^{(\beta)} K_{\gamma\beta} - 1\right]}_{C_\gamma} + \lambda_\delta \underbrace{\left[\sum_{\beta\neq(\delta,\gamma)} \lambda_\beta y_T^{(\delta)} y_T^{(\beta)} K_{\delta\beta} - 1\right]}_{C_\delta} + \underset{(\lambda_\delta,\lambda_\gamma)}{\text{const}}$$

$$\min_{(\lambda_\delta,\lambda_\gamma)} \stackrel{!}{=} \frac{1}{2}\Big[\lambda_\gamma^2 Q_{\gamma\gamma} + \lambda_\delta^2 Q_{\delta\delta} + 2\lambda_\gamma\lambda_\delta Q_{\gamma\delta}\Big] + C_\gamma\lambda_\gamma + C_\delta\lambda_\delta$$

Sequential Minimal Optimization (SMO)

- optimize

$$\min_{(\lambda_\delta,\lambda_\gamma)} \stackrel{!}{=} \frac{1}{2}\left[\lambda_\gamma^2 Q_{\gamma\gamma} + \lambda_\delta^2 Q_{\delta\delta} + 2\lambda_\gamma\lambda_\delta Q_{\gamma\delta}\right] + C_\gamma\lambda_\gamma + C_\delta\lambda_\delta$$

- under the following "box" and "equality" constraints

$$0 \leq \lambda_{\gamma,\delta} \leq \frac{C}{p}\,, \qquad \text{(i)}$$

$$\lambda_\gamma + \underbrace{\frac{y_T^{(\delta)}}{y_T^{(\gamma)}}}_{s}\lambda_\delta = \underbrace{-\frac{1}{y_T^{(\gamma)}}\sum_{\beta\neq(\delta,\gamma)}\lambda_\beta y_T^{(\beta)}}_{d} \quad\Rightarrow\quad \lambda_\gamma + s\lambda_\delta = -d \qquad \text{(ii)}$$

- Analytical solution: Schölkopf and Smola, 2001, p. 308
- Pseudocode: Schölkopf and Smola, 2001, p. 313
- Software: www.csie.ntu.edu.tw/~cjlin/libsvm/
  (also covers multiclass problems, support vector regression, one-class SVMs)

Remarks

Sequential Minimal Optimization (SMO) ...

- ...exploits the fact that the reduced optimization problem can be solved analytically.
- ...needs little memory for the actual update; the storage of the full kernel matrix is quadratic in the number of datapoints.
- ...can be much faster than other optimization methods.
- ...converges fast if the rules to select the λ_i are based on good heuristics

References I

Bernhard Schölkopf and Alexander J Smola. *Learning with kernels: support vector machines, regularization, optimization, and beyond*. MIT press, 2001.

Bernhard Schölkopf, Kah-Kay Sung, Christopher JC Burges, Federico Girosi, Partha Niyogi, Tomaso Poggio, and Vladimir Vapnik. Comparing support vector machines with gaussian kernels to radial basis function classifiers. *IEEE Transactions on Signal Processing*, 45(11): 2758–2765, 1997.

Michael E Tipping. Sparse Bayesian learning and the relevance vector machine. *Journal of Machine Learning Research*, 1(Jun):211–244, 2001.

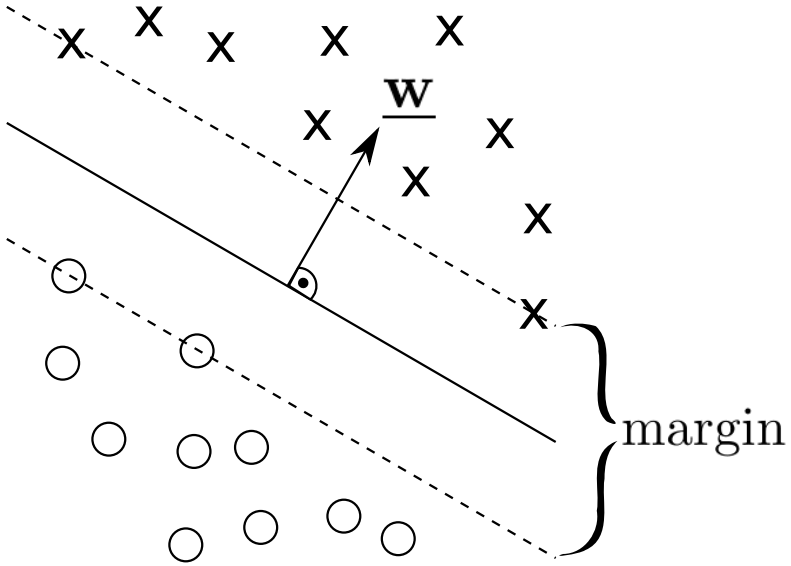
Vladimir Vapnik. *Statistical learning theory*. John Wiley & Sons: Hoboken, 1998.

End of Section 2.2

the following slides contain

OPTIONAL MATERIAL

Classification margin



- **margin**: minimal (nomalized) distance to hyperplane $\mathrm{d}^{\mathsf{min}} = \frac{1}{\|\underline{\mathbf{w}}\|}$
- large margins have low *ambiguity* $\Rightarrow$ low VC-dimension

The solution of the primal problem

Lagrangian

$$L = \frac{1}{2}\|\underline{\mathbf{w}}\|^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \Big\{ y_T^{(\alpha)} \Big(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b \Big) - 1 \Big\}$$

- setting derivative w.r.t. weights w_l to zero: $\underline{\mathbf{w}} = \sum_{\alpha=1}^{p} \lambda_\alpha \, y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)}$

$$\frac{\partial L}{\partial \mathrm{w}_l} = \mathrm{w}_l - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \mathrm{x}_l^{(\alpha)} \stackrel{!}{=} 0$$

The solution of the primal problem

Lagrangian

$$L = \frac{1}{2}\|\underline{\mathbf{w}}\|^2 - \sum_{\alpha=1}^{p} \lambda_\alpha \Big\{ y_T^{(\alpha)} \Big(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b \Big) - 1 \Big\}$$

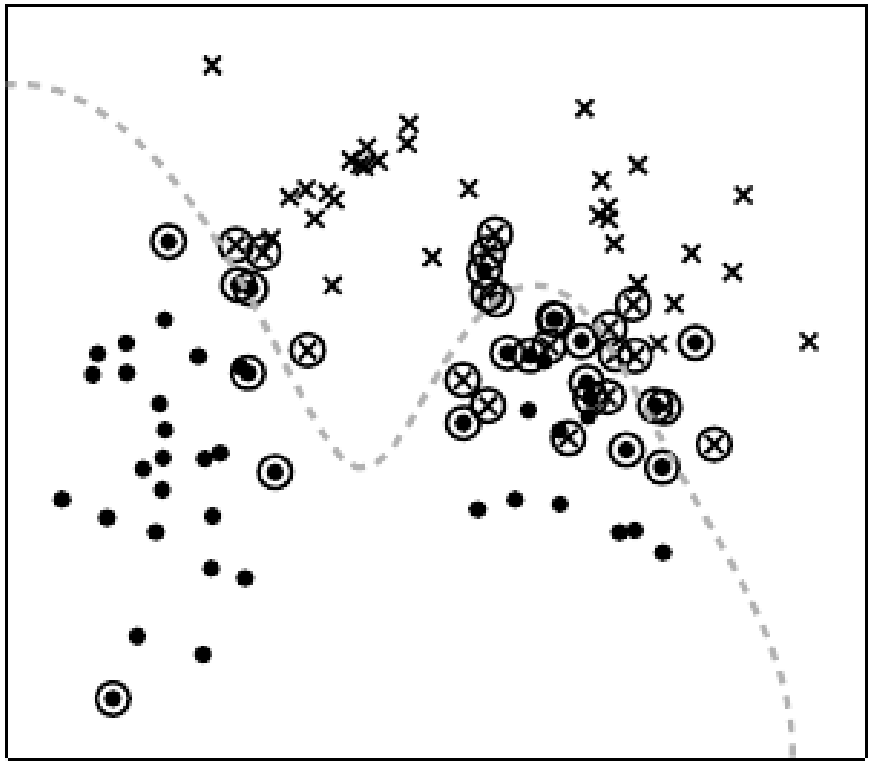
- setting derivative w.r.t. weights w_l to zero: $\underline{\mathbf{w}} = \sum_{\alpha=1}^{p} \lambda_\alpha \, y_T^{(\alpha)} \underline{\mathbf{x}}^{(\alpha)}$

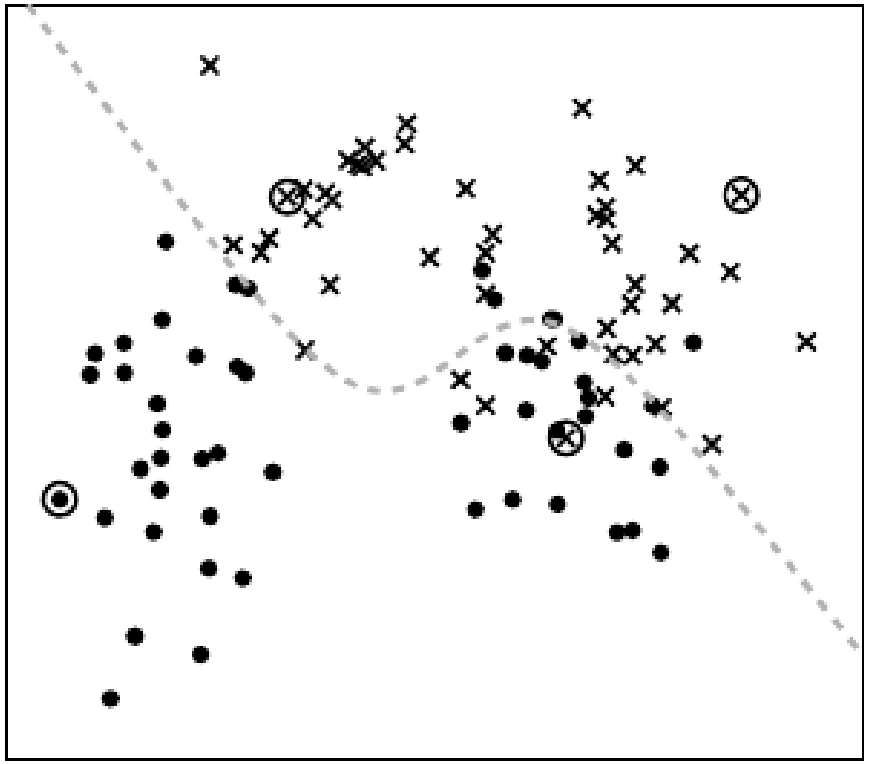
$$\frac{\partial L}{\partial \mathrm{w}_l} = \mathrm{w}_l - \sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \mathrm{x}_l^{(\alpha)} \stackrel{!}{=} 0$$

- setting derivative w.r.t. b to zero

$$\frac{\partial L}{\partial b} = -\sum_{\alpha=1}^{p} \lambda_\alpha y_T^{(\alpha)} \stackrel{!}{=} 0$$

Sparse Bayesian Regression: Relevance Vector Machines





see Tipping (2001)